

★ Square
 Area = ℓ^2
 Perimeter = 4ℓ

★ Rectangle
 Area = $\ell \times b$
 Perimeter = $2(\ell + b)$

★ Cube
 Volume = ℓ^3
 L.S.A = $4\ell^2$
 T.S.A = $6\ell^2$

★ Cuboid
 Volume = $\ell b h$
 L.S.A = $2(\ell b + b h)$
 T.S.A = $2(\ell b + b h + \ell h)$

★ Circle
 Area = πr^2
 Perimeter = $2\pi r$

★ Cylinder
 Volume = $\pi r^2 h$
 C.S.A = $2\pi r h$
 T.S.A = $2\pi r(h + r)$

★ Triangle
 Area = $\frac{1}{2}bh$
 Perimeter = $a + b + c$
 Equilateral Area = $\frac{\sqrt{3}a^2}{4}$
 Equilateral Perimeter = $3a$

★ Cone
 Volume = $\frac{1}{3}\pi r^2 h$
 C.S.A = $\pi r \ell$
 T.S.A = $\pi r(\ell + r)$

★ Herons formula: $A = \sqrt{S(S-a)(S-b)(S-c)}$
 (Semi-perimeter) $S = \frac{a+b+c}{2}$

★ Sphere
 T.S.A = CSA = $4\pi r^2$
 Volume = $\frac{4}{3}\pi r^3$
 hemisphere = TSA = $3\pi r^2$

★ Rhombus Area = $\frac{1}{2}(d_1 \times d_2)$
 Perimeter = $2\sqrt{d_1^2 + d_2^2}$

★ Trapezium Area = $\frac{1}{2}(a+b)h$

★ Sector $\frac{\theta}{360}(\pi r^2)$ [degree]

$\frac{\theta}{180}\pi r$ [radians]

★ Parallelogram Area = $b \cdot c$

Perimeter = $2(b+c)$

★ Hexagon Area = $6 \times \frac{\sqrt{3}}{4} (\text{Side})^2$

★ Percentages

(i) If initial value $\neq$ final value
 $\% \text{ in change } (\uparrow/\downarrow) = \frac{\text{difference}}{\text{Initial value or (then value)}} \times 100$

(ii) If equal, then cycle forms $100 \rightarrow 150$

$$\left\{ \Rightarrow \frac{1}{x} \uparrow \Rightarrow \frac{1}{x+1} \downarrow \right\} \left\{ \Rightarrow \frac{n}{x} \uparrow \Rightarrow \frac{n}{x+n} \downarrow \right\}$$

(iii) Successive $\%$ change

$$\text{Overall } \% \uparrow/\downarrow = a + b + \frac{ab}{100}$$

i.e. $\begin{cases} 1^{\text{st}} \text{ year } 2+ \\ 2^{\text{nd}} \text{ year } 5+ \end{cases}$
 $\rightarrow$ successive

[if $\uparrow \Rightarrow a$, if $\downarrow \Rightarrow -a$]

★ Ratio $A : B \Rightarrow AB : BB : BC$
 $B : C$

$A : B \quad ABC : BBC : BCC : BCD$
 $B : C$
 $C : D$

★ Profit/Loss $SP = CP + P$

$$P/L \% = \frac{SP - CP}{CP} \times 100$$

$$SP = OP \% \cdot CP ; (OP \%) = 100 \% \pm P/L$$

If $SP_1 = SP_2$ & $P \% = L \% = x$ then transaction $n = \frac{-x^2}{100}$

If $CP_1 = CP_2$ & $P \% = L \% \Rightarrow$ no profit/no loss

★ Dishonest Shopkeeper: $P\% = \frac{\text{he gains (gm/kg)}}{\text{for Every how many gm/kg}} \times 100$

Eg: Shopkeeper sell 450gm instead of 500g. $\therefore 50g = \text{gain}$ for every 450 g.

★ Divisibility Rules:

2 $\rightarrow$ last digit should be 0, 2, 4, 6, 8

3 $\rightarrow$ sum of all digits divisible by 3

4 $\rightarrow$ Last two digit divisible by 4 or it is 00

5 $\rightarrow$ Last digit is 0/5

6 $\rightarrow$ divisible by 2 & 3

7 $\rightarrow$ consider last digit & multiply it by 2, Subtract it from rest number.

Repeat it until ~~last~~ digits are divisible by 7 or 0. If not, then it is not divisible by 7

$$\begin{array}{r}
 7546 \\
 -12 \\
 \hline
 742 \\
 -14 \\
 \hline
 70
 \end{array}
 \begin{array}{l}
 \times 2 = 12 \\
 \times 2 = 14 \\
 \text{i.e. divisible by 7}
 \end{array}$$

8 $\rightarrow$ Last 3 digit divisible by 8/000

9 $\rightarrow$ sum of digit should be divisible by 9

(or eliminate 9 & if no number left then number is divisible by 9)

10 $\rightarrow$ number end with 0

11 $\rightarrow$ Sum of odd position - even position should be 0/multiple of 11

12 $\rightarrow$ divisible by 4 & 3

★ L.C.M of fraction = $\frac{\text{L.C.M of Numerator}}{\text{HCF of Denominator}}$

★ HCF of fraction = $\frac{\text{HCF of Numerator}}{\text{LCM of Denominator}}$

★ Shortcut to find LCM = select highest number. ~~take~~ take multiple of it & check it is divisible by other ~~from~~ ^{remaining} numbers or not.

★ HCF = take out common factors of both numbers & multiply it.

① HCF $\left\{ \begin{array}{l} 28 = 2 \times 2 \times 7 \\ 42 = 2 \times 3 \times 7 \end{array} \right. = 2 \times 7 = 14 \text{ is HCF.}$

② LCM (28, 42) = select 42, check 42 is divisible by 28 or not.
 $42 \times 2 = 84$ " " 28 or not, yes
 $\therefore \text{LCM} = 84.$

★ Greatest number dividing a, b, c with leaving same remainder = HCF of $(a-b, b-c, c-a)$

★ Least number with x remainder dividing a, b, c = $\text{LCM of } (a, b, c) + x$

★ Greatest number dividing abc with x, y, z remainder = HCF of $(a-x, b-y, c-z)$

★ Least no. divide a, b, c leave remainder x, y, z = $\text{L.C.M of } (a, b, c) - K$
 where $K = (a-x) \text{ or } (b-y) \text{ or } (c-z)$

★

B	O	D	M	A	S
(), { }, []	of	$\div$	$\times$	$+$	$-$
	(x)				

★ Timed Work revision is Must.

Watch all videos again.

$$\text{Efficiency} \times \text{Days} = \text{Total Marks}$$

Simultaneous Work

$$\begin{cases} C = D \times E \\ D_{A+B} = \frac{C}{E_{A+B}} \\ = \frac{C}{x+y} \end{cases}$$

$$C = LCM(D_1, D_2)$$

$$E_A = \frac{C}{D} = x \text{ c/d}$$

$$E_B = \frac{C}{D} = y \text{ c/d}$$

$$\frac{\text{Man}_1 \text{ Day}_1 \text{ hour}_1}{\text{Work}_1} = \frac{\text{Man}_2 \text{ Day}_2 \text{ hour}_2}{\text{Work}_2}$$

$$M \downarrow \uparrow \propto W \downarrow \uparrow \rightarrow \therefore K = \frac{M_1}{W_1}$$

$$M \downarrow \uparrow \propto H \downarrow \uparrow \rightarrow \therefore K = M_1 H_1$$

A Trigonometry.

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

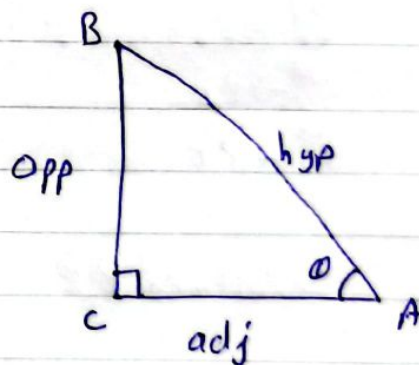
$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\cot \theta = \frac{1}{\tan \theta}$$



$$\cos(A-B) = \cos A \cdot \cos B + \sin A \cdot \sin B$$

$$\sin \theta \cdot \operatorname{cosec} \theta = \cos \theta \cdot \sec \theta = \tan \theta \cdot \cot \theta = 1$$

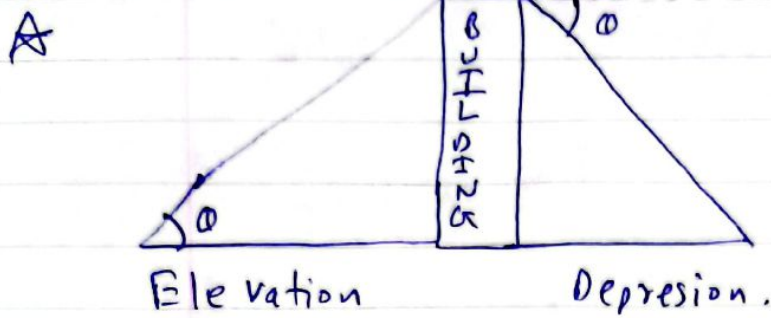
$$\sin^2 \theta + \cos^2 \theta = \sec^2 \theta - \tan^2 \theta = \operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$\begin{aligned} \sin 2x &= 2 \sin x \cos x \\ &= \frac{2 \tan x}{1 + \tan^2 x} \end{aligned}$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\cot 2x = \frac{\cot^2 x - 1}{2 \cot x}$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 2 \cos^2 x - 1 \\ &= 1 - 2 \sin^2 x \\ &= \frac{1 - \tan^2 x}{1 + \tan^2 x} \end{aligned}$$



★ PTR = Simple Interest
100

$$\text{Compound Interest} = P \left[\left(1 + \frac{r}{100} \right)^n - 1 \right]$$

<u>SI</u>	<u>CI</u>	
1 st → Px	Px	1
2 nd → Px	$Px + Px^2$	1 1
3 rd → Px	$Px + 2Px^2 + Px^3$	1 2 1
4 th → Px	$Px + 3Px^2 + 3Px^3 + Px^4$	1 3 3 1

$$\text{Installment } P = \left(1 + \frac{r}{100} \right)^n = x \left(1 + \frac{r}{100} \right)^{n-1} + x \left(1 + \frac{r}{100} \right)^{n-2} + \dots + x \left(1 + \frac{r}{100} \right)^{n-3} + \dots + x$$

$$P = \frac{x}{\left(1 + \frac{r}{100} \right)^1} + \frac{x}{\left(1 + \frac{r}{100} \right)^2} + \dots + \frac{x}{\left(1 + \frac{r}{100} \right)^n}$$

Invest $\times$ time = Profit

Investment ratio = Profit Ratio

Avg = $\frac{\text{Sum}}{n}$

new added person age = $\left(\text{diff. avg} \times \text{initial no. of people} \right) + \text{new avg.}$

$$\begin{aligned}
 \star (a+b)^2 &= a^2 + 2ab + b^2 \\
 (a-b)^2 &= a^2 - 2ab + b^2 \\
 a^2 - b^2 &= (a-b)(a+b) \\
 (a+b+c)^2 &= a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \\
 (a+b)^3 &= a^3 + b^3 + 3ab(a+b) \\
 (a-b)^3 &= a^3 - b^3 - 3ab(a-b) \\
 a^3 + b^3 &= (a+b)(a^2 - ab + b^2) \\
 a^3 - b^3 &= (a-b)(a^2 + ab + b^2) \\
 a^3 + b^3 + c^3 - 3abc &= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) \\
 \text{If } a+b+c &= 0 \text{ then } a^3 + b^3 + c^3 = 3abc
 \end{aligned}$$

★ Rounding off irrational no.

$$\begin{aligned}
 87.\overline{43} &\Rightarrow \text{No.} = \text{Entire no.} - \text{Not repeating.} \\
 &= 8743 - 87 \\
 &= 8656
 \end{aligned}$$

$\overline{99}$ } $\because$ 2 no. repeating $\Rightarrow$ 2 times 9 i.e. 99.

$$1.0\overline{23} \Rightarrow \text{no.} - \text{non repeating.} \quad 1023 - 10 = 1013 \quad \left\{ \begin{array}{l} \text{Numerator} \\ \text{\& denominator} = 99 \end{array} \right.$$

$$\& \text{ denominator} = 99 \quad 0 = 990$$

after decimal 1 non repeating.
after decimal two repeating.

$$\Rightarrow \frac{1013}{990}$$

$$\rightarrow a^p \times b^q \times c^r \Rightarrow \text{factors} = (p+1)(q+1)(r+1)$$

$$\rightarrow \text{Max power of } P \text{ in } N! = \frac{N}{P} + \frac{N}{P^2} + \frac{N}{P^3} + \dots \Rightarrow \sim \text{ans.}$$

$\rightarrow$ If x & y have common factor as only 1 $\Rightarrow$ it is coprime pair.

$$\begin{aligned} \frac{a_1}{a_2}x + \frac{b_1}{b_2}y &= 0 \\ \Rightarrow \frac{a_1}{a_2} &\neq \frac{b_1}{b_2} \Rightarrow \text{unique solution} \end{aligned}$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \Rightarrow \text{infinite solution}$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \text{no solution.}$$

* factorization $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ if $b^2 - 4ac > 0 \rightarrow 2 \text{ diff real roots}$
 $= 0 \rightarrow \text{equal real roots}$
 $< 0 \rightarrow \text{imaginary roots}$

* Avg. Speed = $\frac{\text{Total dist.}}{\text{total time.}}$ $\frac{1 \text{ km}}{\text{hr}} = \frac{5}{18} \frac{\text{m}}{\text{s}}$

$\rightarrow$ When distance is equal then Avg Speed = $\frac{2 \times S_1 S_2}{S_1 + S_2}$

$\rightarrow$ " time is equal " $= \frac{S_1 + S_2}{2}$

$\rightarrow$ Relative speed $\rightarrow$ Same direction / stream = $S_1 - S_2$
 opposite direction / stream = $S_1 + S_2$

$\rightarrow S_{\text{boat}} = \frac{S_{\downarrow \text{stream}} + S_{\uparrow \text{stream}}}{2}$

$\rightarrow S_{\text{stream}} = \frac{S_D - S_U}{2}$

$\rightarrow$ or $\rightarrow +$, and $\rightarrow \times$

$\rightarrow nC_r = \frac{n!}{r! (n-r)!}$ Eg: $10C_2 = \frac{10 \times 9}{2 \times 1}$

$nC_n = 1$	$nC_1 = n$	$nC_{n-1} = n$	$nC_2 = nC_{n-2}$
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→ ${}^n P_r = \frac{n!}{(n-r)!}$ ${}^n P_n = n!$ ${}^n P_1 = n$ ${}^n P_{n-1} = n!$

* repetition not allowed $\Rightarrow n!$

repetition allowed $\Rightarrow$ ~~total unique letters~~ $\times$ ~~no. of blocks~~
 $\frac{n!}{(repeating\ word!) \times (another\ repeating\ word!) \times \dots}$
 ~~2500 = 3333~~

* Sum of number formed = (sum of digit) $\times (n-1)!$ $\times$ 1111
 $\downarrow$
 4 digit no.

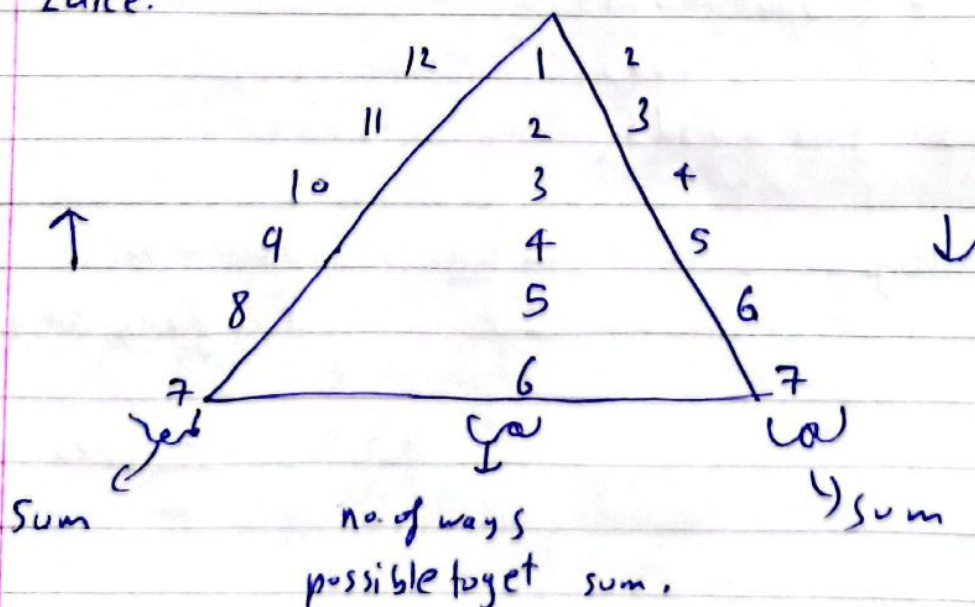
* linear $\rightarrow n!$

Circular $\rightarrow (n-1)!$

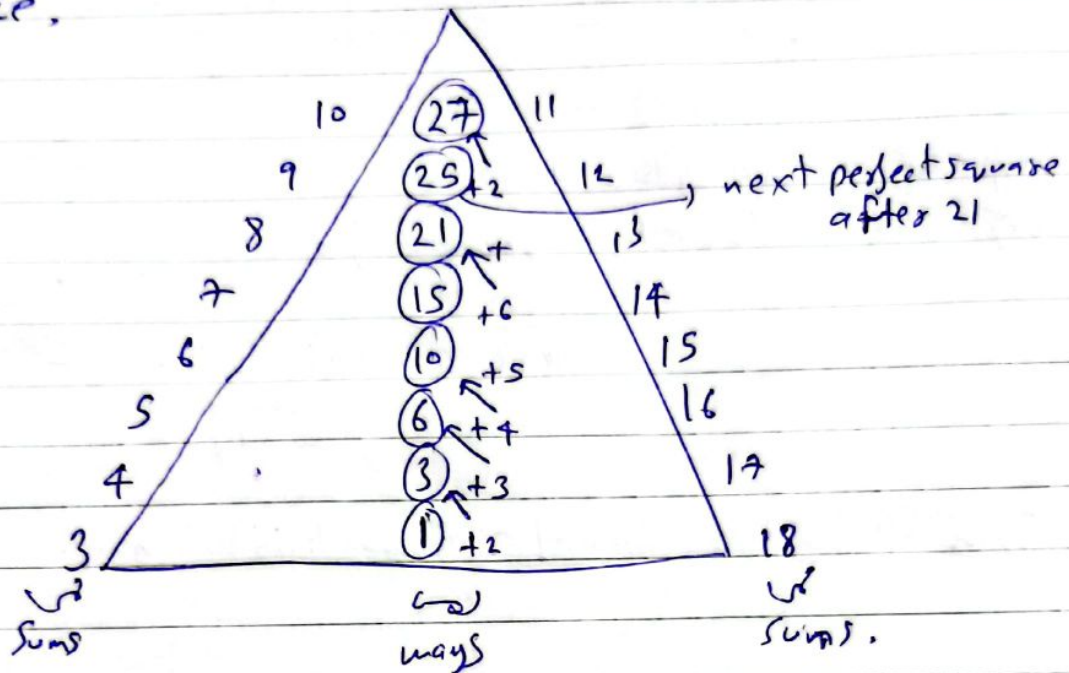
* {necklace, garland} $\Rightarrow \frac{(n-1)!}{2}$

* n coins $\rightarrow 2^n$ possibilities.

* 2 dice.



★ 3 dice.



★ 1 Deck = 52 cards.

Suit 1	Suit 2	Suit 3	Suit 4.
Spade (Black)	Club (Black)	Diamond (Red)	Heart (Red)

2, 3, ... 10 } Spot cards
 A, Q, K, J } Honour cards
 K, Q, J } Face cards.

★ Mean = Avg

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

$x_i = \text{obs}$
 $f = \text{frequency}$

$$x_i = \frac{\text{low} + \text{upper}}{2}$$

★ Median

Ungrouped: odd $\Rightarrow \frac{n+1}{2}$ obs

even $\Rightarrow \frac{\frac{n}{2} \text{ obs} + \frac{n+1}{2} \text{ obs}}{2}$

$$\text{Group} = l + \frac{(\frac{n}{2} - CF)}{f} \times h$$

Eg	Class	f	(f)
	0-10	10	10
	10-20	15	25 = CF
	20-30	12 = f	37 $\uparrow$
	30-40	15	52
	40-50	8	60
		$\frac{60}{2} \Rightarrow 30$	$\Rightarrow 30$ or more i.e. 37 years

★ if 0-9, 10-19, 20-29, 30-39
 $\Rightarrow 0-9.5, 9.5-19.5, 19.5-29.5, \dots$

$$\text{★ Weighted Avg} = \frac{\text{Avg}_1 \times N_{01} + \text{Avg}_2 \times N_{02}}{N_{01} + N_{02}}$$

★ mixture $\frac{x}{1} : \frac{y}{1}$ total 1600 kg.

if 8% x & 12% y $\Rightarrow$ 8% 12%
 (d:10) 11% (d:11)
 12-11 11-8

~~★ Pure = $\left[x \left(1 - \frac{y}{x} \right)^n \right]$~~

★ Pure quantity left in container after diluting milk with water.

x = Pure quantity

y = replaced/removed

n = no. of time process happens.

★ Mode = 3 Median - 2 Mean

★ Mode = $l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$

(f_1 = highest frequency modal class)

★ Range = highest value - lowest value.

★ Mean deviation = $\sum \frac{\text{diff. of range}}{f_i}$

★ S.d = $\sqrt{\text{Variance}}$

Population = S.d. = $\sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$

$\bar{x}$ = mean

x_i = obs.

n = no. of times.

Sample S.D = $\sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

if 2 obs $\Rightarrow \frac{\text{diff}}{2} = \text{S.D.}$

if n natural no. = $\sqrt{\frac{n^2 - 1}{12}}$

★ length of arc = $\frac{\theta}{360} \times 2\pi r$

★ area of arc = $\frac{\theta}{360} \times \pi r^2$

★ Area of sector = $\frac{\text{Arc length} \times \text{Area of circle}}{\text{circumference of circle}}$

- ★ $\theta < 90 \rightarrow$ acute
 $\theta > 90 \rightarrow$ obtuse
 $\theta \geq 180 \rightarrow$ reflex angle.

 vertically opposite.

 corresponding.

 alternate

 angular bisector

complementary = $\theta_1 + \theta_2 = 90^\circ$

Supplementary = $\theta_1 + \theta_2 = 180^\circ$ ($1^\circ = 60'$)

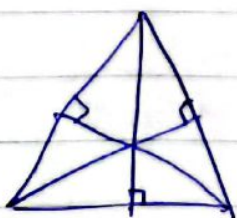
2 sides equal $\rightarrow$ scalen Δ

no side equal $\rightarrow$ isoscles Δ

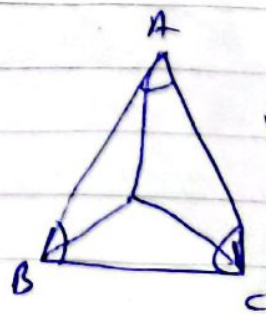
no. of diagonal = $\frac{n(n-3)}{2}$

Sum of int angle = $180(n-2)$

Sum of ext angle = 360 .



median.



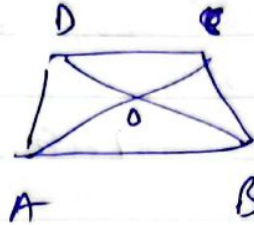
incenter

formed by
angular
bisector

Circle Trigonometry

★

$$\text{Area} = \left(\frac{a+b}{2}\right)h$$



$$\frac{AO}{OC} = \frac{BO}{OD}$$

★

$$n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$$

$$n(X \cup Y \cup Z) = n(X) + n(Y) + n(Z) - n(X \cap Y) - n(Y \cap Z) - n(Z \cap X) + n(X \cap Y \cap Z)$$

$$n(A) = n(A - B) + n(A \cap B)$$

$$n(B) = n(B - A) + n(A \cap B)$$